	Team Members	General Topic	Specific Topic	Comments	Paper Links >
Uras Mutlu (EXAMPLE PROJECT)		Computer Vision	Image to Image Translation with GANs	GANs are widely used in image-to-image translation tasks and there are many successful works. We can reproduce the results of the related papers and we can try to improve the results by applying novel methods.	https://papers.nips.cc/paper/6650-toward-multimodal-image-to-image-translation.pdf http://openaccess.thecvf.com/content_ICCV_2017/papers/Zhu_Unpaired_Image-To-Image_Translation_ICCV_2017_paper.pdf
EXAMPLE PROJECT FROM AI PROJECTS #3		Anomaly Detection	Anomaly detection using generative models like Autoencoders, Variational Autoencoders and GANs.	Generative models are used mostly for computer vision applications. But they are also good at detecting anomalies. They can be trained in an unsupervised way or semi supervised way. By learning the distribution of normal behavior of the system, they can detect anomalous behavior by their reconstruction losses. In this project, we will create an anomaly detection model that detects cyber attack's like DoS, Probing, U2R etc. **Cyber attacks** are our anomalous data in this setting and we will try to detect them by using generative models. (cyber swcurity knowledge is not required)Generative models have promising results in the field of anomaly detection and they are currently state of the art models in this field. We will also compare the performance metrics of these generative models with traditional ones. **Training autoencoder models in semi-supervised and unsupervised way and comparing their results is also within the scope of the this project.** GAN models will be tried if we have enough time. NSL-KDD dataset may be used for this project. P.s.: other type of anomalous data rather than cyber security data can also be used.	https://pdfs.semanticscholar.org/0611/46b1d7938d7a8dae70e3531a00fceb3c78e8.pdf https://ieeexplore.ieee.org/abstract/document/8386760
Onur Boyar	**Onur** Ramazan Oğuz Emre	Recommender Systems	Matrix Factorization, SVD, Funk SVD, Gradient Descent	Building recommender systems without using a library to do matrix computations and optimizations. We will try different approaches and compare them in terms of their performance. **ACM RecSys Challenge 2020**	http://www.recsyschallenge.com/2020/
Hasan Avci	Can Bulguoglu	Computer Vision	Image Recognition & Doorbell	Building a face recognition model and deploy it a Raspberry Pi etc. Then building an architecture to sending SMS when a person detected.	
Hasan Avci		Recommender Systems	LSTM, GRU etc.	A recommendation system for next basket recommendation learning from past purchases. Then challenge it with mf algorithms like NMF	https://cseweb.ucsd.edu/classes/fa17/cse291-b/readi
Uğur Ali Kaplan	**Uğur** Metehan Janberk Hasan Elif	Recommender Systems	User Engagement Prediction	User engagement prediction is a subfield of recommender systems, since companies such as Twitter would like to show you the tweets you are most likely to engage with. The goal is to predict the probability for different types of engagement (Favourite, Reply, Retweet and Quote Tweet) of a target user for a set of tweets, based on heterogeneous input data. This is the **ACM RecSys2020 Challenge**. Details can be read from http://www.recsyschallenge.com/2020/ and if you decide to participate in the challenge, you also have to agree to terms & conditions at http://www.recsyschallenge.com/2020/Terms%20and%20Conditions%20in%20the%20I There are many different methods we could explore. It is possible to use matrix factorization methods, various deep learning algorithms and even reinforcement learning. We can decide on which methods to try after discussing our strong and weak points. We can also rapidly prototype different methods and focus on the most promising, say 2 methods and improve upon them. If we actually end up being successful (Top 5), we will be expected to **write a workshop paper for ACM RecSys 2020 Conference**.	**Survey Paper:** http://bit.ly/2VGchyb **Related Work:** http://bit.ly/2vwKohb http://bit.ly/2PF4aOB http://bit.ly/32Ghh7j http://bit.ly/32DguUM http://bit.ly/3agVs0B http://bit.ly/2VwvL8p
Ulaş Mergen	**Ulaş** Can Büşra	Conversational Assistant	Conversational Information Seeking (CIS).	There are currently few datasets appropriate for training and evaluating models for Conversational Information Seeking (CIS). The main aim of TREC CAsT is to advance research on conversational search systems. The goal of the track is to create a reusable benchmark for open-domain information centric conversational dialogues. The track will run in 2019 and establish a concrete and standard collection of data with information needs to make systems directly comparable. This is the first year of TREC CAsT, which will run as a track in TREC. This year we aim to focus on candidate information ranking in context: Read the dialogue context: Track the evolution of the information need in the conversation, identifying salient information needed for the current turn in the conversation Retrieve Candidate Response Information: Perform retrieval over a large collection of paragraphs (or knowledge base content) to identify relevant information	http://www.treccast.ai/
Batuhan Faik Derinbay	**Batuhan** Atahan Ensar Mehmet Ali	Computer Vision	Robustness of Computer Vision Systems	*Widely available, large datasets *Various pre-made basis models with open source code *Many papers on each dataset and different robustness techniques available for many *There is a suitable upcoming vision challenge in April, due August -Winners gets to present their work in an ECCV 2020 workshop -Challenge link: http://www.robustvision.net/	Note: There are 7 different challenges available with 15 different datasets therefore each challenge and corresponding robustness techniques should be investigated separately. Here is an examplary SotA paper: https://arxiv.org/pdf/1911.06667v1.pdf
Emre Yüksel	**Güneykan** Onur Berk	Anomaly Detection	Anomaly detection using graph neural networks, graph convolutional networks	Graph Neural Network are used for anomaly detection, image and text classification and combinatorial optimization problem. There are many approaches about detecting anomalies in both unsupervised and semi-supervised learning on graph-structured data . We can reproduce the results of the papers and increase the model score by creating new hypotheses and apply them to different datasets.	https://ieeexplore.ieee.org/document/8862186 https://arxiv.org/pdf/2002.12011.pdf https://arxiv.org/pdf/1906.02694.pdf https://arxiv.org/pdf/1609.02907.pdf https://github.com/YingtongDou/graph-fraud-detection-papers Datasets:https://snap.stanford.edu/data/
Guneykan Ozgul		Monte Carlo Filtering	Robot Localization - Object Tracking	One of the well known problem in AI-robotics is to localize a kidnapped robot based on particle filter algorithm. We could implement this solution and try to touch on common issues such as selecting dynamic number of particles and using more sophisticated methods to model the noise and input	https://arxiv.org/pdf/1301.0607.pdf https://arxiv.org/pdf/1911.01383.pdf https://doi.org/10.3182/20020721-6-ES-1901.01132
Batuhan Faik Derinbay		Computer Vision	Image Classification	*Yelp Challenge with many previous work and published papers *NLP (Sentiment analysis) can be used to classify images *Numerious approaches should be applied for classification. E.g.: - Contrast - Depth of Field - Alignment *A photo scoring model should be implemented after classification	https://www.yelp.com/dataset/challenge https://ieeexplore.ieee.org/abstract/document/7484319 https://dl.acm.org/doi/abs/10.1145/2783258.2783380 https://dl.acm.org/doi/abs/10.1145/2806416.2806504
Yağmur Ceren Dardağan	Ceren	Computer Vision	Audio generation with GANs	When dealing with different types of audio (music, speech, sound), generally natural language processing, identification or segmentation of audio come to our minds. We can use generating images with GANs approach for audio generation and investigate the discriminativity of produced audios and the real ones.	https://ieeexplore.ieee.org/abstract/document/8873418 https://paperswithcode.com/paper/convolutional-generative-adversarial-networks https://arxiv.org/abs/1902.08710 https://arxiv.org/abs/1802.04208 Dataset ideas: https://towardsdatascience.com/a-data-lakes-worth-of-audio-datasets-b45b88cd4ad
Berk Sudan		Computer Vision	Image Similarity using Deep Ranking	The ability to find a similar set of images for a given image has multiple use-cases from visual search to duplicate product detection to domain specific image clustering.	[0] https://medium.com/@akarshzingade/image-similarity-using-deep-ranking-c1bd83855978 [1] Improved Consistent Sampling, Weighted Minhash and L1 Sketching- https://static.googleusercontent.com/media/research.google.com/en//pubs/archive/36928.pdf [2] Large scale online learning of image similarity through ranking- http://jmlr.csail.mit.edu/papers/volume11/chechik10a/chechik10a.pdf [3] Learning Fine-grained Image Similarity with Deep Ranking
Halil Eralp Koçaş				I am interested in NVIDIA challenge and Batuhan's Robustness of Computer Vision project.	
Elif Sargut		Recommender Systems	Identifying Product Bundles from Sales Data	We have to employ a clustering technique that is the subjective segmentation to find out the product bundles from sales data.	https://www.researchgate.net/publication/316792527_A_novel_model_for_product_bundling_and_direct_marketing_in_e-
Uras Mutlu	**Uras** **Eralp** **Abdullah** **Can** Ulaş Onur Uğur Janberk	Computer Vision (Social Good Project)	Traffic Anomaly Detection	AI City Challenge Track #4: Detecting anomalies in traffic, including wrong turns, wrong driving direction, lane change errors, and all other anomalies, based on video feeds available from multiple cameras at intersections and along highways.	https://www.aicitychallenge.org/

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Janberk Sü		Text classifier	Nlp	A neural network that filters the comments online regarding the spepicif topic and categorizes,highlights and bundle them to be further analyzed. Getting peoples opinions from social media and preprocessing them to be research is a long process which takes so much time. Yet it is a fruitful, economically feasible field.	
Abdullah BAŞ		Anomaly Detection/Prediction	Brain MRI Anomalies	Metastasis is a major problem in tumors and it causes most of the deaths. Enhancing regions are the main problem that challenges experts and actually it is the major problem/cost of the surgery. There will be a model that can classify sub-types of tumors and predict the lifetime of the patient and also classify enhancing not-tumor type tissue.	https://www.med.upenn.edu/sbia/brats2018/data.html http://braintumorsegmentation.org/
Atahan Özer	Büşra Atahan Güneykan	Unsupervised learning	Using simCLR for audio classification		